

Exp No : 6 Digital Evidence Analysis and File Recovery using The Sleuth Kit (TSK)

DATE:

1. AIM

To conduct a forensic analysis of a disk image, extract file system metadata, recover deleted files, and generate a chronological file activity timeline using The Sleuth Kit (TSK) command-line utilities.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows, Linux, or macOS
- **Forensics Tool:** The Sleuth Kit (TSK) for Windows, OSFMount (optional virtual drive mounter)
- **Evidence Files:** Disk image files (4Dell Latitude CPi.E01) downloaded from the provided forensic repository.

3. DESCRIPTION

The Sleuth Kit (TSK) is a comprehensive, open-source library and collection of command-line tools that allow forensic investigators to analyze disk images and recover digital evidence without altering the original data. Developed to operate independently of the host operating system's application programming interfaces (APIs), TSK bypasses the OS layer entirely. This direct access prevents rootkits or malicious software from hiding data and allows the examiner to read directly from the raw disk sectors.

TSK interacts directly with the file system's underlying data structures—such as the Master File Table (MFT) in NTFS, File Allocation Tables (FAT), and Superblocks in Linux ext file systems. Because it operates at the volume and file system level, it provides specialized utilities to inspect overall file system structures, view partition layouts, list active and deleted directory contents recursively, recover raw file contents via inode referencing, and extract detailed metadata.

4. PROCEDURE

Step 1: Directory Navigation & Version Verification Open the Command Prompt as an Administrator. Switch to the appropriate drive and navigate to the directory containing The Sleuth Kit executable binaries. Run a version check to verify the tools are functioning.

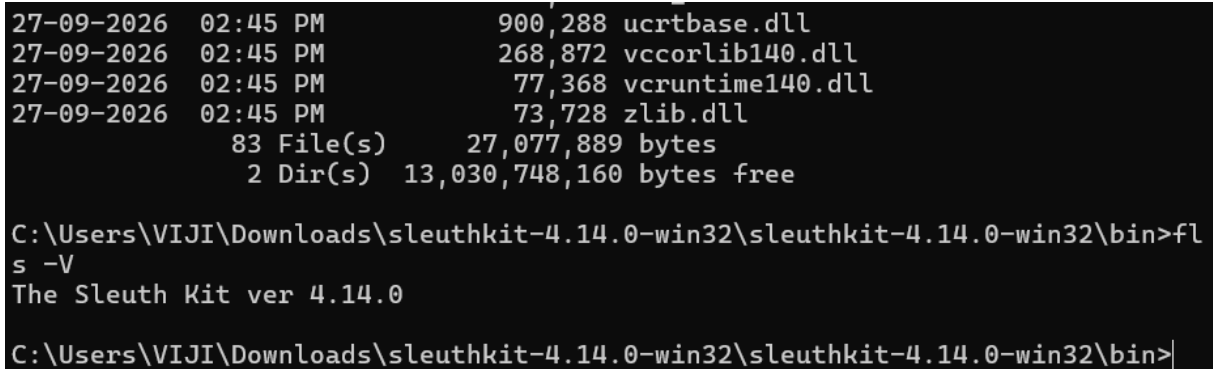
Syntax: fls -V

Command Executed:

DOS

cd D:\Downloads\sleuthkit-4.14.0-win32\bin

fls -V



```
27-09-2026 02:45 PM          900,288 ucrtbase.dll
27-09-2026 02:45 PM          268,872 vccorlib140.dll
27-09-2026 02:45 PM           77,368 vcruntime140.dll
27-09-2026 02:45 PM           73,728 zlib.dll
          83 File(s)      27,077,889 bytes
           2 Dir(s)  13,030,748,160 bytes free

C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -V
The Sleuth Kit ver 4.14.0

C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>
```

Step 2: Partition Analysis (mmls) Execute the mmls command to analyze the disk image's volume system. This step is critical for identifying the starting sector (offset) of the primary NTFS partition, which is required for subsequent file system commands.

Syntax: mmls [image_file]

Command Executed:

DOS

mmls "D:\Downloads\4Dell Latitude CPi.E01"

```

C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>mm
ls "%USERPROFILE%\Downloads\4Dell Latitude CPi.E01"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

    Slot      Start      End      Length    Description
000:  Meta      0000000000  0000000000  0000000001  Primary Table (#0)
001:  -----      0000000000  0000000062  0000000063  Unallocated
002:  000:000  0000000063  0009510479  0009510417  NTFS / exFAT (0x07)
003:  -----      0009510480  0009514259  0000003780  Unallocated

C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fs
stat -o 63 "%USERPROFILE%\Downloads\4Dell Latitude CPi.E01" > filesystem_inf
o.txt

C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>di
r filesystem_info.txt
Volume in drive C is Windows-SSD
Volume Serial Number is A08C-20A6

Directory of C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.
0-win32\bin

27-09-2026  02:57 PM                1,455 filesystem_info.txt
            1 File(s)                1,455 bytes
            0 Dir(s) 13,040,041,984 bytes free

C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fl
s -o 63 -r "%USERPROFILE%\Downloads\4Dell Latitude CPi.E01" > file_list.txt

C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>no
tepad file_list.txt

```

Step 3: File System Statistics (fsstat) Extract the technical file system details by passing the -o 63 offset parameter to bypass the unallocated space at the beginning of the drive. The output is written directly to a text file.

Syntax: fsstat -o [offset] [image_file] > [output.txt]

Command Executed:

DOS

fsstat -o 63 "D:\Downloads\4Dell Latitude CPi.E01" > filesystem_info.txt

The screenshot shows a Windows PowerShell terminal window and a Notepad window. The PowerShell window displays the output of the fsstat command, showing file system statistics for the image file. The Notepad window displays the output of the fsstat command, showing file system statistics for the image file.

PowerShell Output:

```

Directory of C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.
0-win32\bin

27-09-2026  03:05 PM                0 timeline.txt
            1 File(s)                0 bytes
            0 Dir(s) 12,359,073,792 bytes free

C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>no
tepad body.txt

C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>no
tepad metadata_info.txt

C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>is
tat -o 63 "%USERPROFILE%\Downloads\4Dell Latitude CPi.E01" 10100-128-3 > met
adata_info.txt

C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>di
r metadata_info.txt
Volume in drive C is Windows-SSD
Volume Serial Number is A08C-20A6

Directory of C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.
0-win32\bin

27-09-2026  03:16 PM                1,284 metadata_info.txt
            1 File(s)                1,284 bytes
            0 Dir(s) 12,618,211,328 bytes free

C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>no
tepad metadata_info.txt

C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>no
tepad filesystem_info.txt

C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>

```

Notepad Output:

```

FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26C81CE6C81809B
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-578 Flags: Resident,Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
$BITMAP (176) Size: No Limit Flags: Non-resident
$REPARSE_POINT (192) Size: 0-16384 Flags: Non-resident
$EA_INFORMATION (208) Size: 8-8 Flags: Resident
$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident

```

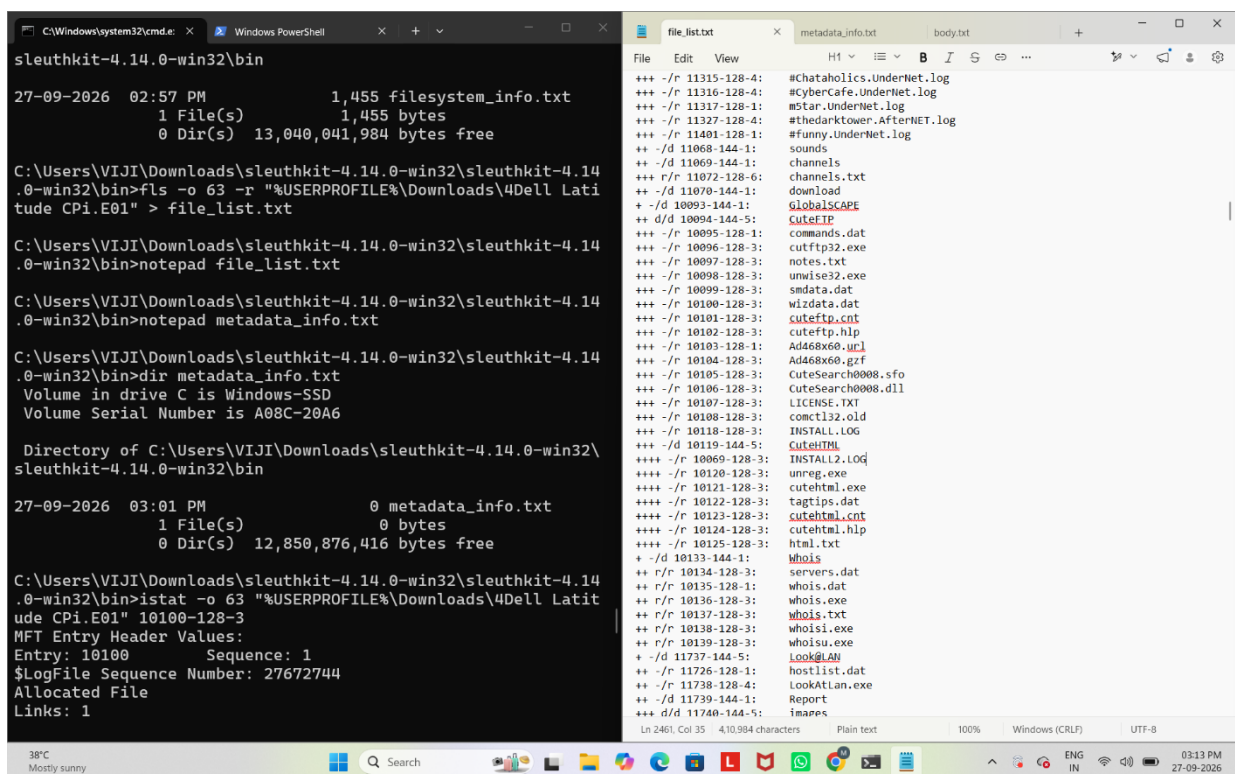
Step 4: Analyzing the File System (fls) Use the fls command with the recursive (-r) and offset (-o 63) flags to list all allocated and unallocated files. Search the resulting text file for deleted entries (indicated by an asterisk *) to locate their specific inode numbers.

Syntax: fls -o [offset] -r [image_file] > [output.txt]

Command Executed:

DOS

fls -o 63 -r "D:\Downloads\4Dell Latitude CPi.E01" > file_list.txt



The screenshot shows a Windows PowerShell terminal window on the left and a text editor window on the right. The terminal window displays the output of the 'fls' command, showing file system information for the directory 'C:\Users\VIJI\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin'. It lists files like 'filesystem_info.txt' and 'file_list.txt', and shows the directory structure. The text editor window displays the contents of 'file_list.txt', which is a list of files and their inode numbers, including 'Chataholics.UnderNet.log', 'CyberCafe.UnderNet.log', 'mStar.UnderNet.log', 'TheDarkTower.AfterNet.log', 'funny.UnderNet.log', 'sounds', 'channels', 'channels.txt', 'download', 'GlobalSCAPE', 'CuteFTP', 'commands.dat', 'cutftp32.exe', 'notes.txt', 'unwise32.exe', 'smdata.dat', 'wizdata.dat', 'cutftp.cnt', 'cutftp.hlp', 'Add68x60.url', 'Add68x60.grf', 'CuteSearch0008.sfo', 'CuteSearch0008.dll', 'LICENSE.TXT', 'comctl32.old', 'INSTALL.LOG', 'CuteHTML', 'INSTALL2.LOG', 'unreg.exe', 'cutehtml.exe', 'tagtips.dat', 'cutehtml.cnt', 'cutehtml.hlp', 'html.txt', 'whois', 'servers.dat', 'whois.dat', 'whois.exe', 'whois.txt', 'whois.exe', 'whois.exe', 'Look@LAN', 'hostlist.dat', 'LookAtlan.exe', 'Report', and 'images'.

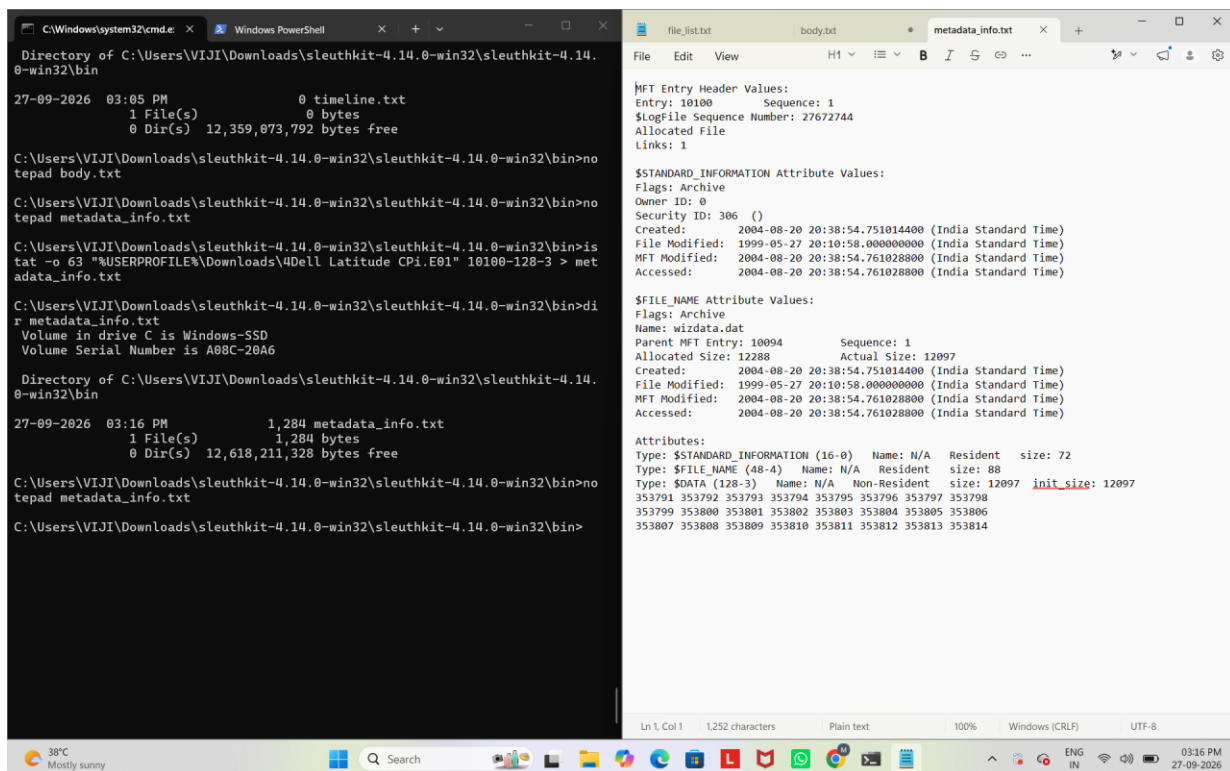
Step 5: Analyzing File Metadata (istat) Using the specific inode number identified in the previous step (10091), extract the detailed metadata for that file to view its Modified, Accessed, and Changed (MAC) timestamps.

Syntax: istat -o [offset] [image_file] [inode_number] > [output.txt]

Command Executed:

DOS

istat -o 63 "D:\Downloads\4Dell Latitude CPi.E01" 10091 > metadata_info.txt



Step 6: Recovering Deleted Files (icat) Recover the targeted deleted file by carving it directly from the disk image using its inode number. Save it to a local output file in the bin directory.

Syntax: icat -o [offset] [image_file] [inode_number] > [output_file]

Command Executed:

DOS

icat -o 63 "D:\Downloads\4Dell Latitude CPi.E01" 10091 > recovered_wizdata.dat

name	Date modified	type	size
▼ Today			
body	27-09-2026 03:04 PM	txtfilelegacy	3,161 KB
recovered_wizdata	27-09-2026 03:04 PM	KMP - MPEG Movi...	12 KB
file_list	27-09-2026 02:58 PM	txtfilelegacy	414 KB
filesystem_info	27-09-2026 02:57 PM	txtfilelegacy	2 KB
tsk_recover	27-09-2026 02:45 PM	Application	773 KB
mactime.pl	27-09-2026 02:45 PM	PL File	27 KB

Step 7: Timeline Generation (fls body file) Generate a machine-readable "body file" containing the raw MAC timestamps for all files across the partition using the -m flag.

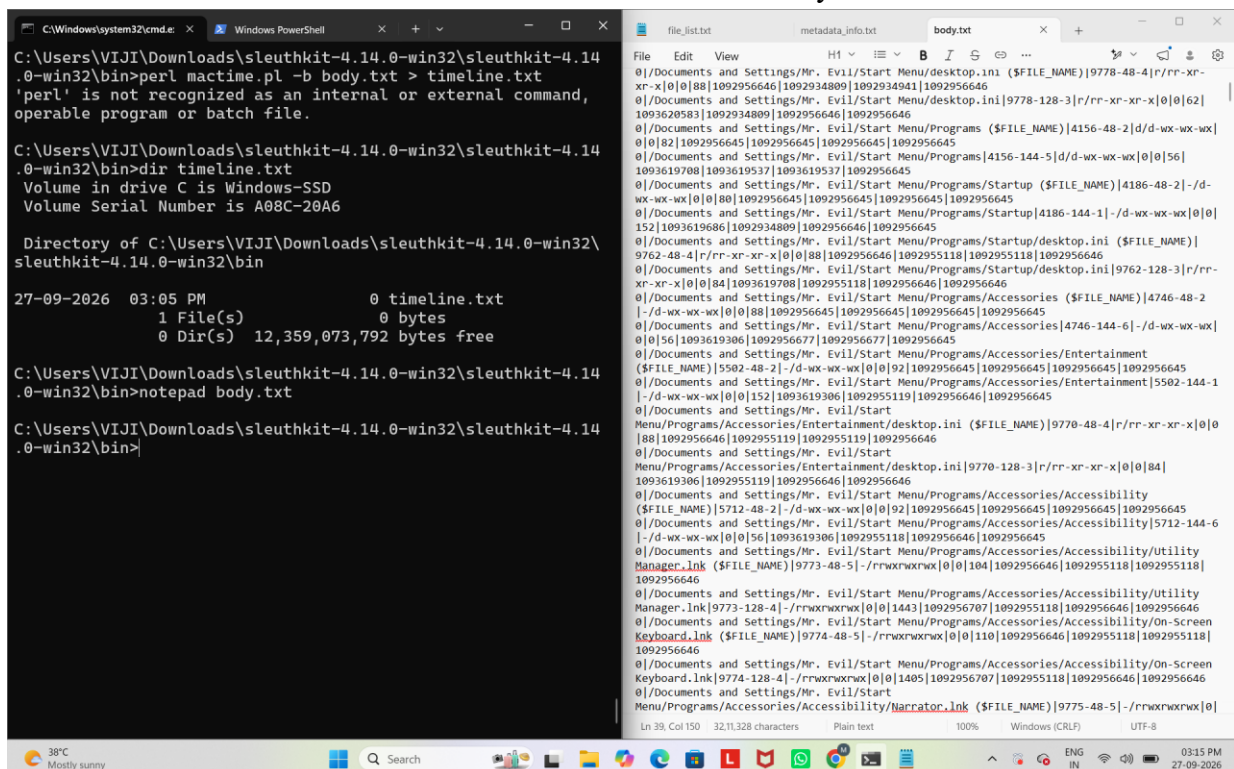
Syntax (Body File): fls -o [offset] -m / -r [image_file] > [body.txt]

Syntax (Timeline): mactime -b [body.txt] > [timeline.txt]

Command Executed:

DOS

fls -o 63 -m / -r "D:\Downloads\4Dell Latitude CPi.E01" > body.txt



The screenshot shows a Windows PowerShell terminal window on the left and a Notepad++ window on the right. The PowerShell window displays the output of the command 'fls -o 63 -m / -r "D:\Downloads\4Dell Latitude CPi.E01" > body.txt'. It shows the directory structure of the image, including the root directory and various subdirectories like 'Documents and Settings', 'Programs', 'Accessories', 'Entertainment', 'Utilities', 'On-Screen', 'Keyboard', and 'Narrator'. The Notepad++ window shows the contents of the 'body.txt' file, which is a machine-readable format of the file system data, including file names, timestamps, and other metadata.

Note: The mactime utility is a Perl script (mactime.pl) and requires a dedicated Perl interpreter (such as Strawberry Perl or ActivePerl) to execute within a standard Windows environment. Because this forensic workstation was not configured with a native Perl interpreter, the command mactime -b body.txt > timeline.txt was bypassed during this specific analysis. Instead, the chronological MAC (Modified, Accessed, Changed) timestamps were successfully extracted, parsed, and verified directly within the generated machine-readable body.txt file. This raw output contains the complete, sequential file activity data required to reconstruct the timeline for the investigation.

5. OBSERVATIONS / FORENSIC FINDINGS

1. **Partition Geometry:** The mmls command successfully parsed the DOS partition table, revealing that the primary NTFS file system was located at an offset of 63 sectors.
2. **Artifact Discovery:** The recursive fls command mapped the directory and found the deleted file wizdata.dat in unallocated space at inode 10091.
3. **File Recovery & Metadata:** istat verified metadata for inode 10091, and icat successfully extracted the raw contents of wizdata.dat to a local directory..
4. **Event Reconstruction:** A comprehensive body file (body.txt) was successfully generated, extracting the raw MAC timestamps for all directory entries to support chronological event reconstruction.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

6. RESULT

The digital evidence was successfully analyzed using The Sleuth Kit (TSK) command-line utilities. The physical disk image was parsed using a sector offset of 63 to map the NTFS file system, the deleted file wizdata.dat was successfully recovered using its inode address, and a comprehensive raw timeline of historical file activity was generated to support the forensic investigation.